LAPORAN PRAKTIKUM

PEMROGRAMAN BERORIENTASI OBJEK 3

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MEMBUAT FUNGSI CRUD SERVICE DAN COSTUMER

1.Membuat table service di database MySQL seperti ini :

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
☐	1	**id**	int(11)			No	*None*		AUTO_INCREMENT
☐	2	**jenis**	varchar(255)	utf8mb4_general_ci		No	*None*		
☐	3	**harga**	double			No	*None*		
☐	4	**status**	varchar(50)	utf8mb4_general_ci		No	*None*		

2.Membuat class ServiceRepo dan codenya seperti ini :

```java
package DAO;

import java.sql.Connection;

public class ServiceRepo implements ServiceDAO {
    private Connection connection;


    final String insert = "INSERT INTO service(id, jenis, harga, status) VALUES(?,?,?,?);";
    final String select = "SELECT * FROM service;";
    final String delete = "DELETE FROM service WHERE id=?;";
    final String update = "UPDATE service SET jenis=?, harga=?, status=? WHERE id=?;";

    public ServiceRepo() {
        connection = Database.koneksi();
    }

    @Override
    public void save(Service service) {
        PreparedStatement st = null;
        try {
            st = connection.prepareStatement(insert);
            st.setString(1, service.getId());
            st.setString(2, service.getJenis());
            st.setString(3, service.getHarga());
            st.setString(4, service.getStatus());
            st.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            try {
                if (st != null) st.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
```

```java
    @Override
    public List<Service> show() {
        List<Service> services = new ArrayList<>();
        Statement st = null;
        ResultSet rs = null;
        try {
            st = connection.createStatement();
            rs = st.executeQuery(select);
            while (rs.next()) {
                Service service = new Service();
                service.setId(rs.getString("id"));
                service.setJenis(rs.getString("jenis"));
                service.setHarga(rs.getString("harga"));
                service.setStatus(rs.getString("status"));
                services.add(service);
            }
        } catch (SQLException e) {
            Logger.getLogger(ServiceDAO.class.getName()).log(Level.SEVERE, null, e);
        } finally {
            try {
                if (rs != null) rs.close();
                if (st != null) st.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
        return services;
    }

    @Override
    public void update(Service service) {
        PreparedStatement st = null;
        try {
            st = connection.prepareStatement(update);
            st.setString(1, service.getJenis());
            st.setString(2, service.getHarga());
            st.setString(3, service.getStatus());
            st.setString(4, service.getId());
```

```java
@Override
public void update(Service service) {
    PreparedStatement st = null;
    try {
        st = connection.prepareStatement(update);
        st.setString(1, service.getJenis());
        st.setString(2, service.getHarga());
        st.setString(3, service.getStatus());
        st.setString(4, service.getId());
        st.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        try {
            if (st != null) st.close();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}

@Override
public void delete(String id) {
    PreparedStatement st = null;
    try {
        st = connection.prepareStatement(delete);
        st.setString(1, id);
        st.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        try {
            if (st != null) st.close();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}
```

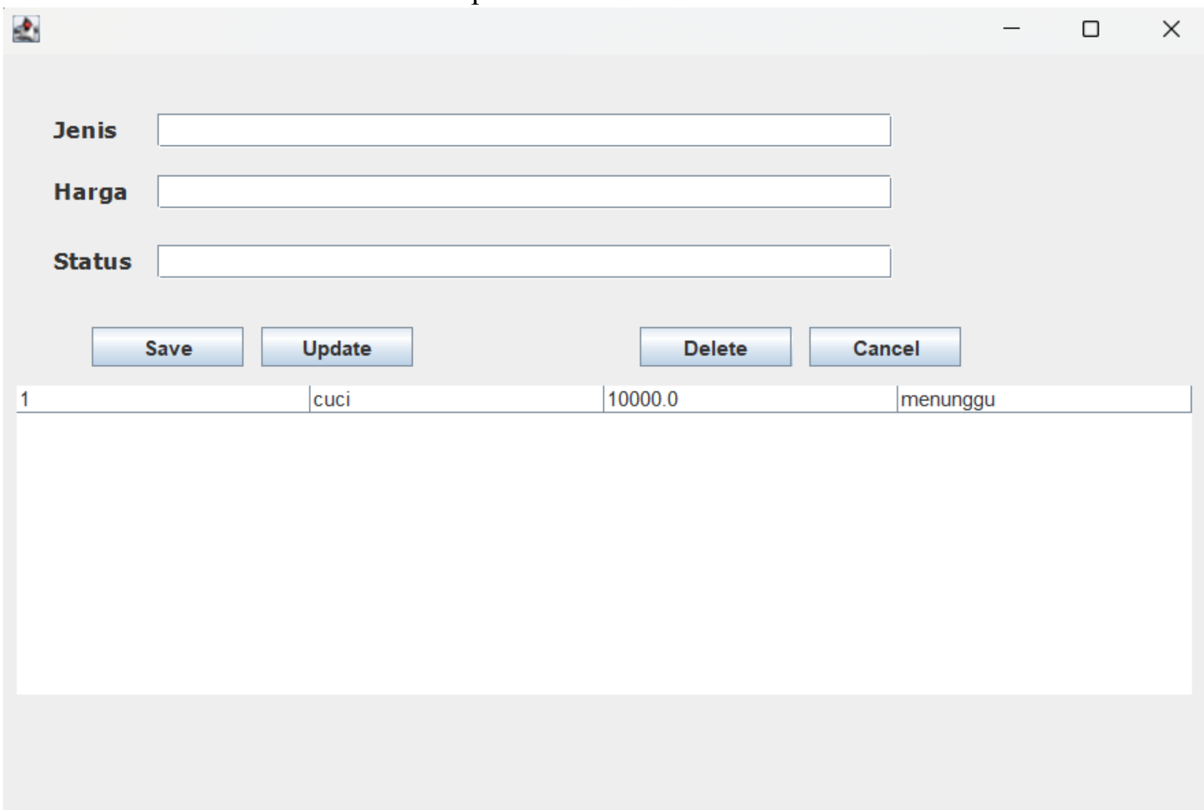
3.Membuat class ServiceDAO dan code seperti ini :

```java
package DAO;

import java.util.List;

public interface ServiceDAO {
    void save(Service service);
    List<Service> show();
    void delete(String id);
    void update(Service service);
}
```

4.Membuat JFrame untuk Service seperti ini :



5.Membuat method reset pada JFrame seperti ini :

```
public void reset() {
    txtJenis.setText("");
    txtHarga.setText("");
    txtStatus.setText("");
}
```

6.Membuat instance pada JFrame ServiceFrame

```
ServiceRepo srvc = new ServiceRepo();
List<Service> ls;
public String id;
```

7.Membuat fungsi CREATE pada ServiceFrame

```
JButton btnSave = new JButton("Save");
btnSave.addActionListener(new ActionListener() {
    public void actionPerformed(ActionEvent e) {
        Service service = new Service();
        service.setJenis(txtJenis.getText());
        service.setHarga(txtHarga.getText());
        service.setStatus(txtStatus.getText());
        srvc.save(service);
        reset();
        loadTable();
    }
});
btnSave.setBounds(54, 160, 89, 23);
contentPane.add(btnSave);
```

8.Membuat fungsi READ pada ServiceFrame

```
public void loadTable() {
    ls = srvc.show();
    TableService tu = new TableService(ls);
    tableService.setModel(tu);
    tableService.getTableHeader().setVisible(true);
}
```

9.Membuat fungsi UPDATE pada ServiceFrame

```
        tableService = new JTable();
        tableService.addMouseListener(new MouseAdapter() {
            @Override
            public void mouseClicked(MouseEvent e) {
                id = tableService.getValueAt(tableService.getSelectedRow(), 0).toString();
                txtJenis.setText(tableService.getValueAt(tableService.getSelectedRow(), 1).toString());
                txtHarga.setText(tableService.getValueAt(tableService.getSelectedRow(), 2).toString());
                txtStatus.setText(tableService.getValueAt(tableService.getSelectedRow(), 3).toString());
            }
        });
        tableService.setBounds(10, 194, 688, 181);
        contentPane.add(tableService);
```

Untuk mengupdate table atau mengubah inputan, ketik code dibawah :

```
        JButton btnUpdate = new JButton("Update");
        btnUpdate.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                Service service = new Service();
                service.setJenis(txtJenis.getText());
                service.setHarga(txtHarga.getText());
                service.setStatus(txtStatus.getText());
                srvc.update(service);
                reset();
                loadTable();
            }
        });
        btnUpdate.setBounds(153, 160, 89, 23);
        contentPane.add(btnUpdate);
```

10.Membuat fungsi DELETE pada ServiceFrame :

```
        JButton btnDelete = new JButton("Delete");
        btnDelete.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                if(id != null) {
                    srvc.delete(id);
                    reset();
                    loadTable();
                }else {
                    JOptionPane.showMessageDialog(null, "Silahkan Pilih Data yang Akan di Hapus");
                }
            }
        });
        btnDelete.setBounds(375, 160, 89, 23);
        contentPane.add(btnDelete);
```

11.Membuat table Costumer di database MySQL seperti ini :

	#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra	Action		
☐	1	**id** 🔑	int(11)			No	*None*		AUTO_INCREMENT	Change	Drop	More
☐	2	**nama**	varchar(128)	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	3	**alamat**	text	utf8mb4_general_ci		No	*None*			Change	Drop	More
☐	4	**nohp**	varchar(15)	utf8mb4_general_ci		No	*None*			Change	Drop	More

12.Membuat class CostumerRepo dan codenya seperti ini :

```java
package DAO;

import java.sql.Connection;

public class CustomerRepo implements CustomerDAO {

    private Connection connection;

    final String insert = "INSERT INTO costumer(nama, alamat, nohp) VALUES(?, ?, ?);";
    final String select = "SELECT * FROM costumer;";
    final String update = "UPDATE costumer SET nama=?, alamat=?, nohp=? WHERE id=?;";
    final String delete = "DELETE FROM costumer WHERE id=?;";

    public CustomerRepo() {
        connection = Database.koneksi();
    }

    @Override
    public void save(Costumer costumer) {
        PreparedStatement st = null;
        try {
            st = connection.prepareStatement(insert);
            st.setString(1, costumer.getNama());
            st.setString(2, costumer.getAlamat());
            st.setString(3, costumer.getNohp());
            st.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            try {
                if (st != null) {
                    st.close();
                }
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
```

```java
@Override
public List<Costumer> show() {
    List<Costumer> ls = new ArrayList<>();
    try {
        Statement st = connection.createStatement();
        ResultSet rs = st.executeQuery(select);
        while (rs.next()) {
            Costumer costumer = new Costumer();
            costumer.setId(rs.getString("id"));
            costumer.setNama(rs.getString("nama"));
            costumer.setAlamat(rs.getString("alamat"));
            costumer.setNohp(rs.getString("nohp"));
            ls.add(costumer);
        }
    } catch (SQLException e) {
        Logger.getLogger(CustomerDAO.class.getName()).log(Level.SEVERE, null, e);
    }
    return ls;
}

@Override
public void update(Costumer costumer) {
    PreparedStatement st = null;
    try {
        st = connection.prepareStatement(update);
        st.setString(1, costumer.getNama());
        st.setString(2, costumer.getAlamat());
        st.setString(3, costumer.getNohp());
        st.setString(4, costumer.getId());
        st.executeUpdate();
    } catch (SQLException e) {
        e.printStackTrace();
    } finally {
        try {
            if (st != null) {
                st.close();
            }
        } catch (SQLException e) {
```

```java
            st.setString(2, costumer.getAlamat());
            st.setString(3, costumer.getNohp());
            st.setString(4, costumer.getId());
            st.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            try {
                if (st != null) {
                    st.close();
                }
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }

    @Override
    public void delete(String id) {
        PreparedStatement st = null;
        try {
            st = connection.prepareStatement(delete);
            st.setString(1, id);
            st.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            try {
                if (st != null) {
                    st.close();
                }
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    }
}
```

13.Membuat class CostumerDAO dan code seperti ini :

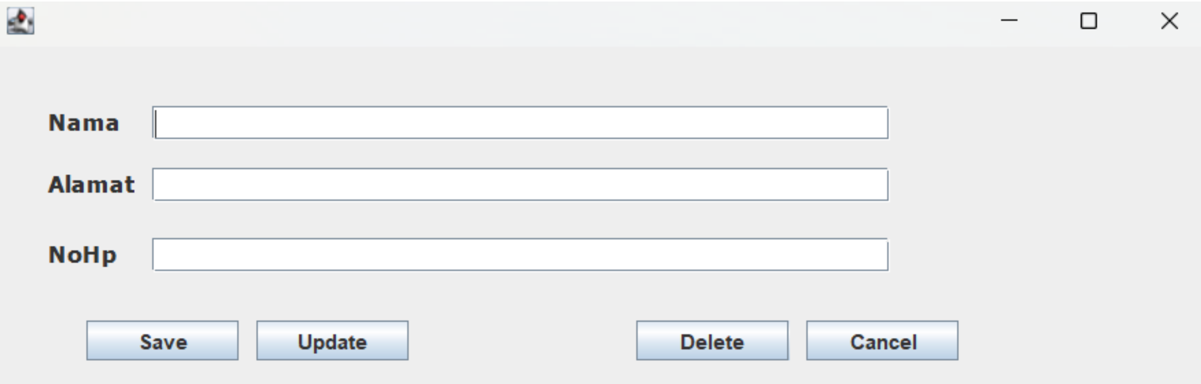
```
package DAO;

import java.util.List;

public interface CustomerDAO {
    void save(Costumer costumer);
    List<Costumer> show();
    void update(Costumer costumer);
    void delete(String id);
}

```

14.Membuat JFrame untuk Costumer seperti ini :



15.Membuat method reset pada JFrame seperti ini :

```
public void reset() {
    txtNama.setText("");
    txtAlamat.setText("");
    txtNohp.setText("");
}
```

16.Membuat instance pada JFrame ServiceFrame

```
CustomerRepo cstmr = new CustomerRepo();
List<Costumer> ls;
public String id;
```

17.Membuat fungsi CREATE pada ServiceFrame

```
        JButton btnSave = new JButton("Save");
        btnSave.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                Costumer costumer = new Costumer();
                costumer.setNama(txtNama.getText());
                costumer.setAlamat(txtAlamat.getText());
                costumer.setNohp(txtNohp.getText());
                cstmr.save(costumer);
                reset();
                loadTable();
            }
        });
        btnSave.setBounds(54, 160, 89, 23);
        contentPane.add(btnSave);
```

18.Membuat fungsi READ pada ServiceFrame

```
    public void loadTable() {
        ls = cstmr.show();
        TableCustomer tu = new TableCustomer(ls);
        tableCostumer.setModel(tu);
        tableCostumer.getTableHeader().setVisible(true);
    }
}
```

19.Membuat fungsi UPDATE pada ServiceFrame

```
        tableCostumer = new JTable();
        tableCostumer.addMouseListener(new MouseAdapter() {
            @Override
            public void mouseClicked(MouseEvent e) {
                id = tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 0).toString();
                txtNama.setText(tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 1).toString());
                txtAlamat.setText(tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 2).toString());
                txtNohp.setText(tableCostumer.getValueAt(tableCostumer.getSelectedRow(), 3).toString());
            }
        });
        tableCostumer.setBounds(10, 194, 688, 228);
        contentPane.add(tableCostumer);
```

```java
    JButton btnUpdate = new JButton("Update");
    btnUpdate.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
            Costumer costumer = new Costumer();
            costumer.setNama(txtNama.getText());
            costumer.setAlamat(txtAlamat.getText());
            costumer.setNohp(txtNohp.getText());
            cstmr.update(costumer);
            reset();
            loadTable();
        }
    });
    btnUpdate.setBounds(153, 160, 89, 23);
    contentPane.add(btnUpdate);
```

20.Membuat fungsi DELETE pada ServiceFrame :

```java
    JButton btnDelete = new JButton("Delete");
    btnDelete.addActionListener(new ActionListener() {
        public void actionPerformed(ActionEvent e) {
            if(id != null) {
                cstmr.delete(id);
                reset();
                loadTable();
            }else {
                JOptionPane.showMessageDialog(null, "Silahkan Pilih Data yang Akan di Hapus");
            }
        }
    });
    btnDelete.setBounds(375, 160, 89, 23);
    contentPane.add(btnDelete);
```

21.Tampilan pada DATABASE :

←T→			id	jenis	harga	status
Edit	Copy	Delete	1	cuci	10000	menunggu

←T→			id	nama	alamat	nohp
Edit	Copy	Delete	1	irfan	padang	02850923852